

QUALICUM BEACH, BC, Canada

WMO: 717660

Lat: 49.340N

Lon: 124.390W

Elev: 58

StdP: 100.63

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.4	-3.3	-9.7	1.7	-2.5	-7.0	2.1	-1.3	8.3	6.1	7.6	5.8	1.4	240	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.0	17.3	25.9	16.7	24.0	16.2	18.2	24.8	17.6	23.7	17.0	22.4	2.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.5	19.7	15.3	11.0	19.2	14.7	10.5	18.8	51.7	25.1	49.6	23.8	47.6	22.6	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	6.0	5.2	DB	-8.6	31.3	1.9	1.9	-9.9	32.6	-11.0	33.7	-12.0	34.8	-13.4	36.1	(4)
(5)			WB	-8.7	19.6	2.0	0.7	-10.2	20.1	-11.3	20.5	-12.5	20.9	-13.9	21.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	3.5	4.1	5.7	8.2	12.6	15.1	18.1	18.0	14.3	9.3	5.7	3.0		
	DBStd	6.02	2.91	2.80	2.41	2.30	2.86	2.75	2.34	2.30	2.57	2.41	3.16	3.05		
	HDD10.0	964	202	166	134	63	8	0	0	0	1	42	132	216		
	HDD18.3	3160	460	399	391	306	179	104	31	34	123	279	379	474		
	CDD10.0	905	0	0	1	8	88	153	252	248	131	21	3	0		
	CDD18.3	60	0	0	0	0	0	8	25	24	3	0	0	0		
	CDH23.3	544	0	0	0	0	14	77	219	206	28	0	0	0		
	CDH26.7	116	0	0	0	0	1	18	52	43	3	0	0	0		
(14) Wind	WSAvg	2.0	2.0	2.0	2.3	2.1	1.9	1.9	1.9	1.7	1.7	1.9	2.2	2.2		
(15) Precipitation	PrecAvg	1085	171	102	98	61	45	42	22	34	41	116	167	171		
	PrecMax	1346	320	326	210	130	88	82	42	117	134	260	339	235		
	PrecMin	771	87	6	20	19	13	11	3	3	9	19	57	86		
	PrecStd	165	62	75	47	32	21	19	12	28	31	58	83	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.7	15.0	19.0	25.2	29.1	31.1	30.1	26.6	18.2	14.1	11.5		
		MCWB	8.7	9.3	10.5	11.6	14.8	16.7	18.7	17.8	17.2	13.6	11.5	9.5		
	2%	DB	9.5	10.2	12.1	15.8	22.8	25.7	28.0	27.8	23.4	16.5	12.6	9.3		
		MCWB	8.1	7.8	8.7	9.6	14.1	16.1	17.6	17.3	15.7	12.4	10.7	8.0		
	5%	DB	8.4	9.1	10.7	14.1	20.4	22.4	25.7	25.7	21.3	14.7	11.4	8.2		
		MCWB	7.3	7.2	7.9	9.3	13.4	15.1	16.8	17.0	15.3	11.8	9.9	7.1		
	10%	DB	7.5	8.1	9.7	12.6	18.2	20.2	23.8	23.6	19.4	13.5	10.2	7.2		
		MCWB	6.5	6.7	7.4	8.7	12.9	14.3	16.2	16.6	14.5	11.2	9.0	6.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.3	11.3	13.1	15.9	18.1	19.6	19.1	18.4	15.0	12.5	10.1		
		MCDB	10.2	11.2	13.9	16.4	22.3	25.0	28.1	25.5	24.0	17.2	13.5	11.1		
	2%	WB	8.4	8.7	9.5	11.1	14.9	16.8	18.2	18.2	17.1	13.6	11.1	8.3		
		MCDB	9.1	9.4	11.4	14.4	20.5	23.6	25.1	24.7	21.6	15.2	12.1	9.0		
	5%	WB	7.5	7.7	8.6	10.1	14.1	15.8	17.5	17.6	16.0	12.4	10.2	7.2		
		MCDB	8.2	8.5	10.1	12.5	19.2	21.0	23.8	23.8	20.1	13.9	11.1	8.0		
	10%	WB	6.6	6.9	7.8	9.3	13.3	14.9	16.9	17.0	15.1	11.4	9.2	6.4		
		MCDB	7.3	7.8	9.3	11.6	17.4	19.2	22.4	22.3	18.2	12.9	10.1	7.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.9	6.9	8.1	9.1	8.6	10.0	10.0	8.9	7.2	6.0	5.0		
		MCDBR	5.5	6.6	7.9	10.5	13.6	12.9	14.1	13.7	12.9	9.0	6.2	5.1		
	5% WB	MCWBR	4.0	4.6	5.1	5.8	6.1	5.0	5.4	5.3	6.3	5.7	4.5	4.1		
		MCWBR	5.0	5.1	6.9	8.3	11.0	10.5	11.8	11.4	10.4	7.5	5.5	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.307	0.327	0.340	0.363	0.356	0.346	0.347	0.332	0.332	0.314	0.309		
		taud	2.511	2.581	2.534	2.491	2.443	2.480	2.551	2.554	2.584	2.555	2.544	2.484		
	Ebn at Noon		749	858	892	910	897	903	908	893	875	811	745	700		
		Edh at Noon	60	70	85	99	108	105	96	92	81	71	57	54		
(44) All-Sky Solar Radiation	RadAvg		0.84	1.74	2.58	4.02	5.20	5.62	6.14	5.24	3.71	2.01	1.00	0.66		
	RadStd		0.12	0.27	0.34	0.39	0.53	0.62	0.58	0.43	0.32	0.31	0.18	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page